Purav Bhatt

Full-Stack & AI Engineer | TypeScript, Node.js, Next.js, Python, Flutter, AWS, LLMs/RAG

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Full-stack and AI engineer shipping production systems end-to-end: a workforce platform in daily use (Flutter, Next.js, Supabase), a chaos-tested job queue (Node.js, BullMQ, Redis), and RAG/LLM features on Amazon Bedrock deployed to AWS with CDK and CI/CD.

Education

Guru Gobind Singh Indraprastha University — Delhi, India
Bachelor of Technology in Artificial Intelligence and Data Science — *Expected 2027*

Experience

Full-Stack Developer — Sep 2025 – Present
City Fire Services India — *Delhi, India*

- Built and shipped Marka, a production workforce platform used daily by field crews: two Flutter apps and a Next.js/TypeScript admin portal on Supabase; 1,100+ commits via PR-based review and GitHub Actions CI.
- Designed the data layer: PostgreSQL with Row-Level Security on every table, REST APIs, Supabase Edge Functions (Deno) and pg_cron jobs, with secrets in Supabase Vault.
- Built live GPS tracking (Mapbox), on-device OCR (Google ML Kit), biometric check-in, FCM push and payroll; fixed GPS race conditions with cancellable async fetches.

Data Science Intern — Jul 2025 – Aug 2025
IIIT Delhi — *Delhi, India*

- Built feature pipelines and scikit-learn baselines, and ran graph neural network (GNN) training pipelines for the research team.

Projects

Marka | *Flutter, Riverpod, Next.js, Supabase, PostgreSQL, Mapbox* | live

- Attendance and payroll platform: GPS-verified check-in, live crew map, OCR expenses, leave approvals and auto-calculated salary slips.

PulseBoard | *Next.js, Node.js, Socket.io, PostgreSQL, pgvector, Redis, AWS Bedrock, CDK* | live

- Real-time collaborative Kanban (100 concurrent WebSocket users, 78.5 ms p95), deployed to AWS ECS Fargate, RDS and CloudFront with CDK and GitHub Actions over OIDC.
- RAG "Ask the board" with cited answers and hybrid search (pgvector + full-text, Reciprocal Rank Fusion): Recall@5 0.957, citation precision 1.0, 10/10 prompt injections resisted.
- LLM evals that gate CI on regressions, AI triage validated with zod, and an XGBoost completion-time model (MAE 16.7 h vs. 26.7 h baseline).

RelayQ | *TypeScript, Node.js, BullMQ, Redis, PostgreSQL, Docker, OpenTelemetry, AWS* | live

- Multi-tenant job service: 33,424 jobs/min; 0 lost or duplicated across 10k jobs with 10 worker kills, via a transactional outbox and idempotent effect log.
- LLM jobs with per-tenant budgets, fallback models and a cost ledger; retries, dead-letter queue, rate limits, HMAC webhooks; traced with OpenTelemetry.

Urban Policy Simulation | *Python, Mesa, OSMnx, FastAPI, Next.js*

- Team project: built the agent-based simulation core on Delhi's real road network with multi-modal routing and a calibrated BPR congestion model.

Technical Skills

Languages: TypeScript, JavaScript, Python, Dart, SQL **Frameworks**: React, Next.js, Node.js, Express, Flutter, Riverpod, FastAPI, Tailwind CSS, Socket.io **Data**: PostgreSQL, Prisma, Redis, BullMQ, Supabase, pgvector **AI & ML**: LLMs, RAG, embeddings, semantic search, LLM evals, Amazon Bedrock, Anthropic API, XGBoost, scikit-learn, pandas, GNNs **Cloud & DevOps**: AWS (ECS Fargate, RDS, ElastiCache, S3, CloudFront, CDK), Docker, GitHub Actions, OpenTelemetry, Prometheus, Vitest, Playwright